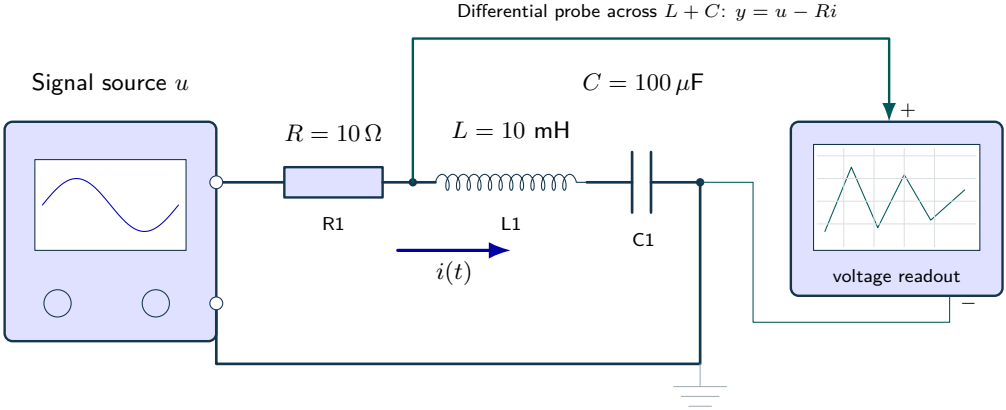

Ideal series network; the output port includes direct feedthrough

Electrical states

Current i [A] and capacitor voltage v_C [V] store magnetic and electric energy.

Circuit equations

$L\dot{i} = u - Ri - v_C$.
$C\dot{v}_C = i$.
Output: $y = u - Ri = v_L + v_C$.

Measurement boundary

The two probe leads span $L + C$. The depicted instrument display is illustrative, not a measured response.